FIT Lecture # 4

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Number Systems:

- Decimal system
- Binary system
- Hexadecimal system
- Octal system

DECIMAL SYSTEM

- The decimal system is the most common number system used by human beings. It is a positional number system that uses 10 as a base to represent different values. Therefore, this number system is also known as base10 number system. In this system, 10 symbols are available for representing the values. These symbols include the digits from 0 to 9. The common operations performed in the decimal system are addition (+), subtraction (–), multiplication (×) and division (/).

BINARY SYSTEM

- The following are some of the technical terms used in binary system:
- Bit. It is the smallest unit of information used in a computer system. It can either have the value 0 or 1. Derived from the words Binary digit.
- Nibble. It is a combination of 4 bits.
- Byte. It is a combination of 8 bits. Derived from words ‘by eight’.
- Word. It is a combination of 16 bits.
- Double word. It is a combination of 32 bits.
- Kilobyte (KB). It is used to represent the 1024 bytes of information.
- Megabyte (MB). It is used to represent the 1024 KBs of information.
- Gigabyte (GB). It is used to represent the 1024 MBs of information.

Binary point

1 0 1 0 0 1 . 0 1 0 1

In binary system, the point used to separate the integer and the fraction part of a number is known as *binary point*. Table 6.2 lists the weights associated with each bit in the given binary number.

Table 6.2 Place values in binary system

Digit	1	0	1	0	0	1	.	0	1	0	1
Weight	2^5	2^4	2^3	2^2	2^1	2^0		2^{-1}	2^{-2}	2^{-3}	2^{-4}

Like the decimal system, the powers to the base increases by 1 towards the left for the integer part and decreases by 1 towards the right for the fraction part. The value of the given binary number can be determined as the sum of the products of the bits multiplied by the weight of the bit itself. Therefore, the value of the binary number 101001.0101 can be obtained as:

$$1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} + 1 \times 2^{-4}$$

$$= 32 + 8 + 1 + 0.25 + 0.0625$$

$$= 41.3125$$

The binary number 101001.0101 represents the decimal value 41.3125.

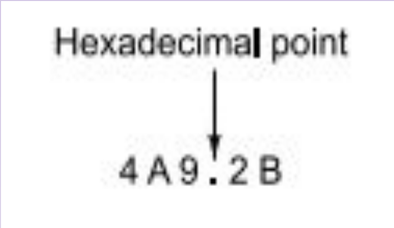
Table 6.3 lists the 4-bit binary representation of decimal numbers 0 through 15.

Table 6.3 Binary representation of first 16 numbers

Decimal number	4-bit binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

HEXADECIMAL SYSTEM

- The hexadecimal system is a positional number system that uses base 16 to represent different values. Therefore, this number system is known as base-16 system. As this system uses base 16, 16 symbols are available for representing the values in this system. These symbols are the digits 0–9 and the letters A, B, C, D, E and F. The digits 0–9 are used to represent the decimal values 0 through 9 and the letters A, B, C, D, E and F are used to represent the decimal values 10 through 15.
- The weight associated with each symbol in the given hexadecimal number can be determined by raising 16 to a power equivalent to the position of the digit in the number. To understand this concept, let us consider the following hexadecimal number:



HEXADECIMAL SYSTEM

In hexadecimal system, the point used to separate the integer and the fraction part of a number is known as hexadecimal point. Table 6.4 lists the weights associated with each digit in the given hexadecimal number.

Table 6.4 Place values in hexadecimal system

Digit	4	A	9	.	2	B
Weight	16^2	16^1	16^0		16^{-1}	16^{-2}

The value of the hexadecimal number can be computed as the sum of the products of the symbol multiplied by the weight of the symbol itself. Therefore, the value of the given hexadecimal number is:

$$4 \times 16^2 + 10 \times 16^1 + 9 \times 16^0 + 2 \times 16^{-1} + 11 \times 16^{-2}$$
$$= 1024 + 160 + 9 + 0.125 + 0.0429$$
$$= 1193 + 0.1679$$
$$= 1193.1679$$

The hexadecimal number 4A9. 2B represents the decimal value 1193.1679.

OCTAL SYSTEM

- The octal system is the positional number system that uses base 8 to represent different values. Therefore, this number system is also known as base-8 system. As this system uses base 8, eight symbols are available for representing the values in this system. These symbols are the digits 0 to 7.
- The weight associated with each digit in the given octal number can be determined by raising 8 to a power equivalent to the position of digit in the number. To understand this concept, let us consider the following octal number:

OCTAL SYSTEM

Octal point

↓

2 1 5 . 4 3

In octal system, the point used to separate the integer and the fraction part of a number is known as *octal point*. Table 6.5 lists the weights associated with each digit in the given octal number.

Table 6.5 Place values in octal system

Digit	2	1	5	.	4	3
Weight	8^2	8^1	8^0		8^{-1}	8^{-2}

Using these place values, we can now determine the value of the given octal number as:

$$2 \times 8^2 + 1 \times 8^1 + 5 \times 8^0 + 4 \times 8^{-1} + 3 \times 8^{-2}$$
$$= 128 + 8 + 5 + 0.5 + 0.0469$$
$$= 141 + 0.5469$$
$$= 141.5469$$

The octal number 215.43 represents the decimal value 141.5469.

Table 6.6 lists the octal representation of decimal numbers 0 through 15.

OCTAL SYSTEM:

Table 6.6 Octal representation of first 16 numbers

Decimal number	Octal representation
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	10
9	11
10	12
11	13
12	14
13	15
14	16
15	17

6.6 4-BIT BINARY CODED DECIMAL (BCD) SYSTEMS

The BCD system is employed by computer systems to encode the decimal number into its equivalent binary number. This is generally accomplished by encoding each digit of the decimal number into its equivalent binary sequence. The main advantage of BCD system is that it is a fast and efficient system to convert the decimal numbers into binary numbers as compared to the pure binary system. However, the implementation of this coding system requires a lot of circuits that makes the design of the computer systems very complicated.

The 4-bit BCD system is usually employed by the computer systems to represent and process numerical data only. In the 4-bit BCD system, each digit of the decimal number is encoded to its corresponding 4-bit binary sequence. The two most popular 4-bit BCD systems are:

- Weighted 4-bit BCD code
- Excess – 3 (XS – 3) BCD code

6.6.1 Weighted 4-Bit BCD Code

The weighted 4-bit BCD code is more commonly known as 8421 weighted code. It is called weighted code because it encodes the decimal system into binary system by using the concept of positional weighting into consideration. In this code, each decimal digit is encoded into its 4-bit binary number in which the bits from left to right have the weights 8, 4, 2, and 1 respectively. Table 6.7 lists the weighted 4-bit BCD code for decimal digits 0 through 9.

Table 6.7 Weighted 4-bit BCD codes

Decimal digits	Weighted 4-bit BCD code
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

Note: *Apart from 8421, some other weighted BCD codes are 4221, 2421 and 5211.*

Example 6.1 *Represent the decimal number 5327 in 8421 BCD code.*

Solution

The given decimal number is 5327.
The corresponding 4-bit 8421 BCD representation of decimal digit 5 is 0101.
The corresponding 4-bit 8421 BCD representation of decimal digit 3 is 0011.
The corresponding 4-bit 8421 BCD representation of decimal digit 2 is 0010.
The corresponding 4-bit 8421 BCD representation of decimal digit 7 is 0111.
Therefore, the 8421 BCD representation of decimal number 5327 is 0101 0011 0010 0111.

Example 6.2 *Convert the decimal number 159 to 8421 BCD code.*

Solution

The given decimal number is 159.
The corresponding 4-bit 8421 BCD representation of decimal digit 1 is 0001.
The corresponding 4-bit 8421 BCD representation of decimal digit 5 is 0101.
The corresponding 4-bit 8421 BCD representation of decimal digit 9 is 1001.
Therefore, the 8421 BCD representation of decimal number 159 is 0001 0101 1001.

Example 6.3 *Convert the decimal number 87. 34 to 8421 BCD code.*

Solution

The given decimal number is 87.34.
The corresponding 4-bit 8421 BCD representation of decimal digit 8 is 1000.
The corresponding 4-bit 8421 BCD representation of decimal digit 7 is 0111.
The corresponding 4-bit 8421 BCD representation of decimal digit 3 is 0011.
The corresponding 4-bit 8421 BCD representation of decimal digit 4 is 0100.
Therefore, the 8421 BCD representation of decimal number 87.34 is 1000 0111. 0011 0100.

Example 6.4 *Determine the decimal number corresponding to the 8421 BCD code 01001001.*

Solution

The given 8421 BCD code is 01001001.

To determine the equivalent decimal number, simply divide the 8421 BCD code into sets of 4-bit binary digits as:

0100 1001

The decimal number corresponding to the binary digits 0100 is 4.

The decimal number corresponding to the binary digits 1001 is 9.

Therefore, the decimal number equivalent to 8421 BCD code 0100 1001 is 49.

Example 6.5 *Determine the decimal number corresponding to the 8421 BCD code 0111010000100110.*

Solution

The given 8421 BCD code is 0111010000100110.

To determine the equivalent decimal number, simply divide the 8421 BCD code into sets of 4-bit binary digits as:

0111 0100 0010 0110

The decimal number corresponding to binary digits 0111 is 7.

The decimal number corresponding to binary digits 0100 is 4.

The decimal number corresponding to binary digits 0010 is 2.

The decimal number corresponding to binary digits 0110 is 6.

Therefore, the decimal number equivalent to 8421 BCD code 0111010000100110 is 7426.

Example 6.6 *Determine the decimal number corresponding to the 8421 BCD representation 010100100001.0110.*

Solution

The given 8421 BCD code is 010100100001.0110.

To determine the equivalent decimal number, simply divide the 8421 BCD code into sets of 4-bit binary digits as:

0101 0010 0001 . 0110

The decimal number corresponding to binary digits 0101 is 5.

The decimal number corresponding to binary digits 0010 is 2.

The decimal number corresponding to binary digits 0001 is 1.

The decimal number corresponding to binary digits 0110 is 6.

Therefore, the decimal number equivalent to 8421 BCD representation 010100100001.0110 is 521.6.

6.6.2 Excess-3 BCD Code

The Excess-3 (XS-3) BCD code does not use the principle of positional weights into consideration while converting the decimal numbers to 4-bit BCD system. Therefore, we can say that this code is a non-weighted BCD code. The function of XS-3 code is to transform the decimal numbers into their corresponding 4-bit BCD code. In this code, the decimal number is transformed to the 4-bit BCD code by first adding 3 to all the digits of the number and then converting the excess digits, so obtained, into their corresponding 8421 BCD code. Therefore, we can say that the XS-3 code is strongly related with 8421 BCD code in its functioning. Table 6.8 lists the XS-3 BCD code for decimal digits 0 through 9.

Table 6.8 Excess-3 BCD codes

Decimal digits	Excess-3 BCD code
0	0011
1	0100
2	0101
3	0110
4	0111
5	1000
6	1001
7	1010
8	1011
9	1100

Note: *Apart from XS-3 code, the other non-weighted BCD code is 4-bit Gray code.*

Example 6.7 *Convert the decimal number 85 to XS-3 BCD code.*

Solution

The given decimal number is 85.

Now, add 3 to each digit of the given decimal number as:

8 + 3 = 11

5 + 3 = 8

The corresponding 4-bit 8421 BCD representation of the decimal digit 11 is 1011.

The corresponding 4-bit 8421 BCD representation of the decimal digit 8 is 1000.

Therefore, the XS-3 BCD representation of the decimal number 85 is 1011 1000.

Example 6.8 *Represent the decimal number 173 in XS-3 BCD code.*

Solution

The given decimal number is 173.

Now, add 3 to each digit of the given decimal number as:

1 + 3 = 4

7 + 3 =10

3 + 3 = 6

The corresponding 4-bit 8421 BCD representation of the decimal digit 4 is 0100.

The corresponding 4-bit 8421 BCD representation of the decimal digit 10 is 1010.

The corresponding 4-bit 8421 BCD representation of the decimal digit 6 is 0110.

Therefore, the XS-3 BCD representation of the decimal number 173 is 0100 1010 0110.

Example 6.9 *Convert the decimal number 3456 into XS-3 BCD code.*

Solution

The given decimal number is 3456.

Now, add 3 to each digit of the given decimal number as:

$3 + 3 = 6$

$4 + 3 = 7$

$5 + 3 = 8$

$6 + 3 = 9$

The corresponding 4-bit 8421 BCD representation of decimal digit 6 is 0110.

The corresponding 4-bit 8421 BCD representation of decimal digit 7 is 0111.

The corresponding 4-bit 8421 BCD representation of decimal digit 8 is 1000.

The corresponding 4-bit 8421 BCD representation of decimal digit 9 is 1001.

Therefore, the XS – 3 BCD representation of decimal number 3456 is 0110 0111 1000 1001.

Note: *4-bit BCD systems are inadequate for representing and handling non-numeric data. For this purpose, 6-bit BCD and 8-bit BCD systems have been developed.*


Thank You